Superposed Bateman Biexponentials with Clearance-Constrained Scaling and Per-User Calibration:

A Pharmacokinetic Framework for Hormone Replacement Therapy Monitoring in the xT Mobile Application

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Abstract

We present the pharmacokinetic (PK) model underlying xT, a mobile application that predicts serum hormone concentration over time from a user's logged dose history for gender-affirming hormone replacement therapy (HRT). The model employs a one-compartment Bateman biexponential—first-order absorption from an intramuscular depot into a single well-mixed compartment, first-order elimination from it—with linear superposition of all prior dose contributions, a dynamic endogenous baseline that decays under gonadal suppression, and a per-user calibration factor α derived from self-reported blood tests. Ten medication/route/form combinations are supported: estradiol (valerate, cypionate, oral, transdermal) and testosterone (cypionate, enanthate, propionate, undecanoate, oral, transdermal).

The scaling constant S—an empirical surrogate for F/V_d—is derived in closed form for testosterone cypionate from a primary population-PK source (Bi et al. 2018), and for the remaining injectable testosterone esters by a conservation-of-mass identity that constrains S to depend only on the elimination rate k_e and testosterone clearance, independent of the ester's half-life. Estradiol ester constants are anchored to compiled steady-state PK tables. Transdermal routes are modeled as 24 hourly micro-doses approximating square-wave release.

We report accuracy against published anchors (0.0–0.1% for anchored quantities), out-of-sample validation (1–15% for estradiol and testosterone single-dose and steady-state checks), and time-to-peak ($t_{\max}$) agreement with primary and secondary sources. We quantify structural limitations: testosterone undecanoate's peak/trough ratio (8.1× modeled vs. 1.8× observed, a source conflict rather than a structural deficiency), transdermal intraday ripple ($C_{\max}/C_{\min} \approx 1.05$ vs. 2.24 observed), and oral testosterone peak fluctuation under-prediction. The model is honest about where its constants are primary-sourced, where they are secondary, and where they remain uncalibrated.

Keywords: pharmacokinetics, Bateman function, hormone replacement therapy, testosterone cypionate, estradiol valerate, flip-flop kinetics, superposition, per-user calibration, mobile health

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1 Introduction

Gender-affirming hormone replacement therapy (HRT) involves the administration of exogenous hormones—estradiol and/or testosterone—to align an individual's serum hormone levels with their gender identity. Clinical management relies on periodic blood draws to measure serum hormone concentration, titrate dosage, and confirm therapeutic levels [10]. Between draws, patients and providers have limited tools to anticipate where levels are heading, when a peak or trough is expected, or how a missed dose or regimen change will affect the trajectory.

xT is a mobile application that addresses this gap by predicting serum hormone concentration over time from the user's logged dose history. It implements a pharmacokinetic model that computes, for any point in time, the superposed contribution of every dose the user has taken, plus a dynamic endogenous baseline, scaled by an optional per-user calibration factor derived from blood test results.

The model covers ten medication/route/form combinations across two hormone classes (Table 1). Its mathematical core is the Bateman biexponential—the analytic solution to one-compartment first-order absorption and elimination—which the published pharmacometric literature independently identifies as the best-fit model for every injectable testosterone ester examined: cypionate [1], enanthate [2], and undecanoate [5]. This convergence is not incidental; it reflects the fact that depot testosterone esters, across a wide range of chain lengths and dosing intervals, produce terminal phases that are single-exponential on semi-log plots, justifying one-compartment fits.

This paper describes the model in full: its governing equation, the provenance and derivation of every constant, the calibration mechanism, the accuracy of its predictions against published data, and the structural limitations that bound what it can honestly claim. It is written to be checkable: every constant is stated with its source, and every derivation is given in enough detail to reproduce.

1.1 Contributions

1. A complete description of a pharmacokinetic model designed for consumer HRT tracking, including the full constant set, per-form derivations, and the dynamic endogenous baseline model.
2. A conservation-of-mass identity (Section 2.8) that constrains the scaling constant S for all injectable testosterone esters to depend only on k_e and clearance, eliminating S as a free parameter and enforcing mutual consistency across esters.
3. An analytic closed-form derivation of S for testosterone cypionate from primary population-PK parameters (Section 2.7), which serves as the anchor for the identity above.
4. A per-user calibration framework ($\alpha = C_{\text{measured}}/C_{\text{predicted}}$) that closes part of the between-subject variability gap without requiring a formal mixed-effects model.
5. An honest accounting of structural limitations, including a source-conflict diagnosis for testosterone undecanoate's peak/trough ratio that reframes a previously assumed structural deficiency as a calibration problem.

2 Methods

2.1 System overview

xT predicts serum hormone concentration over time from a user's logged dose history. Two medication classes are supported, each across three administration routes, yielding ten medication/route/form combinations (Table 1).

Route pseudo-esters. The two non-injectable cases in each ester enumeration describe a *route*, not a chemical ester. They are resolved from the dose record's administration method by a route-

Medication	Route	Ester / form	Output unit
Estradiol	Injectable	Valerate (EV), Cypionate (EC)	pg/mL
Estradiol	Oral	(route pseudo-ester)	pg/mL
Estradiol	Cream	(route pseudo-ester)	pg/mL
Testosterone	Injectable	Cypionate (TC), Enanthate (TE), Propionate (TP), Undecanoate (TU)	ng/dL
Testosterone	Oral	undecanoate (route pseudo-ester)	ng/dL
Testosterone	Cream	unesterified gel (route pseudo-ester)	ng/dL

Table 1: The ten medication/route/form combinations the model covers.

to-ester mapping function, never chosen by the user directly. This design ensures that a dose logged as "oral testosterone" uses the oral route's bioavailability and rate constants, not those of whatever injectable ester was previously selected.

Exogenous and endogenous separation. The PK engine returns the *exogenous* contribution only—the sum of all administered dose contributions. Endogenous production is modeled separately as a dynamic baseline $C_{\text{base}}(t)$ that decays as gonadal suppression progresses (Section 2.13). This separation matters for interpreting several constants, most importantly testosterone cypionate's half-life: the value encoded (3.84 d) is the exogenous-only disposition rate, not the total-testosterone terminal half-life of ∼7 d that is commonly quoted and that includes endogenous secretion.

2.2 Governing equation

Each administered dose contributes a Bateman biexponential—first-order absorption from a depot into a single well-mixed compartment, first-order elimination from it:

$$C_i(t) = D_i \cdot \text{EDF} \cdot S \cdot \frac{k_a}{k_a - k_e}\left(e^{-k_e t} - e^{-k_a t}\right), \qquad t \geq 0 \tag{1}$$

where D_i is the administered dose in mg, EDF is the effective dose factor (Section 2.5), S is a scaling constant with units of concentration per mg, and k_a, k_e are rate constants in day^{-1}. Total predicted level at time t is the superposition of all doses taken before it, plus the endogenous baseline, times a user calibration factor:

$$C(t) = \alpha \left[C_{\text{base}}(t) + \sum_{i\,:\,t_i \leq t} C_i(t - t_i) \right] \tag{2}$$

Superposition is exact here because Equation 1 is linear in dose. That linearity is also the model's principal simplification: there is no saturable protein binding, no SHBG dynamics, no dose-dependent clearance.

The $k_a \approx k_e$ limit. When $|k_a - k_e| < 10^{-4}$, Equation 1 is singular, and the engine substitutes its limit,

$$D \cdot \text{EDF} \cdot S \cdot k_a\, t\, e^{-k_a t},$$

which is the standard degenerate-case form of the Bateman function when the two rate constants coincide.

Transdermal route. The cream/gel route does not use Equation 1. It is modeled as 24 hourly micro-doses of $D/24$ each, decaying single-exponentially ($k_e = 0$), approximating a 24-hour square-wave release:

$$C_{\text{transdermal}}(t) = \sum_{h=0}^{23} \frac{D}{24} \cdot S \cdot e^{-k_a(t-t_h)}, \qquad t \geq t_h \tag{3}$$

where t_h is the time of the h-th micro-dose. This construction has no absorption phase; Section 5.1 explains why it under-models intraday variation.

2.3 Time to peak

Differentiating Equation 1 with respect to t and setting the derivative to zero gives the time to peak concentration:

$$t_{\max} = \frac{\ln(k_a/k_e)}{k_a - k_e} \tag{4}$$

A property of Equation 4 that has practical consequences is its *symmetry* under $k_a \leftrightarrow k_e$: swapping the two rate constants leaves $t_{\max}$ unchanged. For depot esters the two exponentials are not separately identifiable from clinical data, so which one is labelled "absorption" and which "elimination" is a modelling convention rather than a finding. The observable curve is unchanged by the swap, with the amplitude difference absorbed into S.

2.4 Flip-flop kinetics

For depot esters, absorption from the oil vehicle is slower than elimination of the released hormone ($k_a \ll k_e$). The terminal slope of the curve is then governed by *absorption*, not elimination— "flip-flop" kinetics [11]. The practical consequence is that the parameter controlling the terminal half-life sets k_a and governs the entire inter-dose peak-to-trough decay, while the parameter controlling the fast exponential sets k_e and primarily determines where the peak falls.

In xT's parameterization, the field `halfLifeDays` sets $k_a = \ln 2/t_{1/2,\text{term}}$ (the slow rate), and `eliminationHalfLifeHours` sets $k_e = \ln 2/(t_{1/2,\text{elim}}/24)$ (the fast rate). The naming is historical; the doc comments in the codebase explicitly note the convention.

2.5 Effective dose factor

EDF = EWF $\times$ F converts administered mg of ester into systemically available mg of active hormone.

2.5.1 Ester weight factor

The ester weight factor is pure stoichiometry—the mass fraction of the ester molecule that is the parent hormone:

$$\text{EWF} = \frac{M_{\text{hormone}}}{M_{\text{ester}}}$$

with $M_{\text{E}_2} = 272.4$ and $M_{\text{T}} = 288.4\,\text{g/mol}$. These are the only constants in the entire model that are not fitted to anything and cannot be wrong. Table 2 gives the values for all esters.

2.5.2 Bioavailability

F is route bioavailability—the fraction of the administered dose reaching systemic circulation:

Oral testosterone undecanoate carries $F = 1.0$ not because it is fully bioavailable—it is not—but because its absolute bioavailability is not established in the source. Since F and S are confounded (only their product is identified by the published data), folding bioavailability into S avoids inventing a number. Predictions are unaffected by where the split is drawn.

Ester	Abbrev.	EWF	Molecular basis
Estradiol valerate	EV	0.764	272.4/356.5
Estradiol cypionate	EC	0.687	272.4/396.6
Estradiol (oral/TD)	E2	1.000	base hormone, no ester
Testosterone cypionate	TC	0.699	288.4/412.6
Testosterone enanthate	TE	0.720	288.4/400.6
Testosterone propionate	TP	0.837	288.4/344.5
Testosterone undecanoate	TU	0.631	288.4/456.7
Testosterone (gel)	T	1.000	base hormone, no ester

Table 2: Ester weight factors from molecular weights. Oral estradiol and transdermal routes deliver the base hormone directly, so EWF = 1.

Route	F	Basis
Injectable depot	1.0	Near-complete absorption from oil depot
Oral estradiol	0.05	First-pass metabolism
Transdermal E2	0.15	Skin barrier limits absorption
Transdermal T	0.10	AndroGel 1% label §12.3 (stated verbatim)
Oral TU	1.0	Not established; folded into S (see text)

Table 3: Route bioavailability values.

2.6 The scaling constant

S is an empirical surrogate for F/V_d. It is the one constant with no direct physical interpretation, and it is the constant most sensitive to calibration errors. Wherever the sources allow, S is derived in closed form rather than by curve fit; the derivations are given in Sections 2.7 and 2.8.

2.7 Testosterone cypionate: primary-source derivation

Testosterone cypionate (TC) is the only form in the model derived from a primary population-PK publication: Bi et al. [1], who report $\mathrm{CL}/F = 2.6\,\mathrm{kL/day}$, $V/F = 14.4\,\mathrm{kL}$, and $K_a = 1.22\,\mathrm{day}^{-1}$ (Table 2 of the paper, with reported relative standard errors and bootstrap confidence intervals).

2.7.1 Rate constants

The disposition rate is $\mathrm{CL}/V = 2.6/14.4 = 0.1806\,\mathrm{day}^{-1}$, giving a half-life of 3.839 d, which sets k_a (the slow rate under flip-flop). $K_a = 1.22$ gives $\ln 2/1.22 = 13.64\,\mathrm{h}$, which sets k_e (the fast rate). Per the symmetry noted after Equation 4, the labels are conventional; the observable curve is unchanged by the swap, with the amplitude difference absorbed into S.

2.7.2 Why 3.84 days and not 7–8

The commonly quoted TC half-life is around a week. Bi et al. address this discrepancy directly: noncompartmental analysis of their own data yields 6.9 d, and their model yields 6.87 d *when endogenous secretion is assumed to be zero*—but 4.05 d once endogenous production is accounted for. Because xT models endogenous production separately as $C_{\mathrm{base}}(t)$, the exogenous-only figure is the one that belongs in the ester constant. Bi's own median post-hoc estimate of 4.05 d agrees with the encoded 3.839 d to 5.2%.

2.7.3 Scaling constant in closed form

Matching Equation 1 to the one-compartment model underlying Bi's parameters gives S analytically rather than by curve fit:

$$S = \frac{10^5 \cdot K_a}{V_L \cdot \mathrm{EDF} \cdot k_e} = \frac{10^5 \cdot 1.22}{14400 \times 0.699 \times 0.180556} = 67.13 \tag{5}$$

The 10^5 converts mg/L to ng/dL; EDF appears because the app multiplies the dose by it while Bi's V/F already absorbs ester conversion. This reproduces Bi's population model exactly.

2.7.4 Time to peak

Bi's parameters give $t_{\max} = 1.84\,\mathrm{d}$ via Equation 4. Propagating the published 95% CI on K_a (0.785–2.67) gives a band of 1.08–2.43 d. Bi's own limitations section (1–2 absorption-phase samples per subject, K_a RSE 44.9%) explains the imprecision. Read TC $t_{\max}$ as $1.84 \pm 0.7\,\mathrm{d}$.

2.8 Clearance identity for TE, TP, and TU

Testosterone enanthate (TE), propionate (TP), and undecanoate (TU) lack primary population-PK sources with reported V/F and K_a for the intramuscular route in the model's target population. Their scaling constants are nonetheless *not* free parameters, because of a conservation argument that follows from the TC derivation.

2.8.1 Derivation

Integrating Equation 1 over all time gives exactly

$$\int_0^\infty \frac{k_a}{k_a - k_e}\left(e^{-k_e t} - e^{-k_a t}\right)\,dt = \frac{k_a}{k_a - k_e}\left(\frac{1}{k_e} - \frac{1}{k_a}\right) = \frac{1}{k_e}$$

so at steady state with dosing interval τ, superposition gives

$$C_{\mathrm{avg}} = \frac{D \cdot \mathrm{EDF} \cdot S}{\tau\, k_e} \tag{6}$$

The true steady-state average is (mg of testosterone per day) divided by clearance—and clearance is a property of *testosterone*, not of the ester carrying it. It is therefore already pinned by the TC refit: $\mathrm{CL}_T = 2600 \times \mathrm{EWF}_{TC} = 1817.4\,\mathrm{L/day}$. Equating the two expressions, the ester weight factor cancels:

$$\boxed{S = \frac{10^5 \cdot k_e}{\mathrm{CL}_T \cdot F}} \tag{7}$$

S depends only on k_e and clearance. The ester's half-life (k_a) does not appear at all—a property worth stating plainly, because it means S does not track the ester's terminal half-life but rather its fast rate constant, which for depot esters under flip-flop is the absorption rate from the oil vehicle.

2.8.2 The load-bearing check

TC satisfies Equation 7 to 0.03% *without having been fit to it*—its S came from Equation 5, via V/F and K_a. Two independent derivations landing on the same number is what makes it defensible to apply the identity to the other three esters.

2.8.3 Application to TE, TP, and TU

Table 4 shows the identity applied. TE and TU now carry real absorption rates from primary sources (Section 2.9), while TP retains a placeholder 2 h k_e and is flagged as uncalibrated.

Ester	k_e (d^{-1})	S (encoded)	S (implied)	Error
Cypionate	1.2196	67.13	67.11	0.03%
Enanthate	1.9625	107.98	107.98	0.00%
Propionate	8.3178	457.67	457.67	0.00%
Undecanoate	0.3748	20.62	20.62	0.00%

Table 4: The clearance identity (Equation 7) applied to all injectable testosterone esters. $CL_T = 1817.4\,L/day$, $F = 1.0$ for all.

2.8.4 Steady-state predictions at typical regimens

Ester	Regimen	C_{avg}	C_{max}	C_{min}
TC	100 mg q7d	550	736	321
TE	100 mg q7d	566	703	405
TP	50 mg q2d	1151	1743	760
TU	1000 mg q70d	496	1045	129

Table 5: Steady-state predictions (exogenous only, ng/dL) at typical regimens. C_{max} and C_{min} are the peak and trough of the steady-state oscillation.

2.9 Testosterone enanthate and undecanoate: absorption rates

2.9.1 Testosterone enanthate (TE)

TE's absorption rate is derived from Ichihara et al. [2], the only identified primary IM pharmacokinetic study in transgender men: $n = 10$, single 250 mg IM dose (Enarmon Depot). The paper reports per-subject absorption rate constants (k_a) in the 4 of 10 subjects where it was estimable: 2.82, 3.16, 0.49, 1.38 day^{-1}, giving a mean of 1.9625 day^{-1} ($t_{1/2} = 8.48$ h). The terminal half-life is reported as 6.6 ± 1.9 d, which sets k_a (the slow rate) at $\ln 2/6.6 = 0.1050\,day^{-1}$.

The resulting $t_{max} = 1.58$ d agrees with the paper's observed 1.7 ± 0.9 d (all subjects peaked within 3 days). The t_{max} and the k_a values were estimated in different subsets (Tmax in all 10, k_a in the 4 where it was estimable), so this is a genuine cross-check rather than a round trip.

Magnitude discrepancy. A single 250 mg dose gives 881 ng/dL exogenous here against Ichihara's observed C_{max} of 2940 ± 1710 ng/dL. The gap is cohort-level, not error: Ichihara's own CL_{tot}/F (14.3 L/day/kg $\approx$ 930 L/day at 65 kg) is about half the $CL_T = 1817.4$ L/day inherited from Bi, and their V_d/F (118.7 L/kg $\approx$ 7700 L) about half Bi's 14 400 L. Adopting Ichihara's clearance for TE would put TE and TC 2× apart on the *same molecule*, breaking the conservation-of-mass identity. Magnitude is therefore anchored to Bi; only shape was taken from Ichihara. The per-user calibration factor α exists to absorb individual-level differences of this magnitude, and Ichihara's C_{max} CV of 0.58 confirms that between-subject spread is wide enough to accommodate the gap.

2.9.2 Testosterone undecanoate (TU)

TU's terminal half-life (20.0 d) is retained from Zhang et al. [3], who measured 18.3 ± 2.3 d (500 mg) and 23.7 ± 2.7 d (1000 mg); 20.0 sits inside both arms. The absorption rate is fitted so that t_{max} lands on the Aveed (FDA) clinical pharmacology median of 7 d (range 4–42 d): solving Equation 4 with $k_a = \ln 2/20.0 = 0.0347\,day^{-1}$ for k_e gives $k_e = 0.3748\,day^{-1}$ ($t_{1/2} = 44.4$ h), and $t_{max} = 7.00$ d.

The scaling constant follows from the identity: $S = 10^5 \times 0.3748/1817.4 = 20.62$.

Peak/trough ratio. Over a 70-day interval, the model produces a peak/trough ratio of 8.1× (1045/129 ng/dL). Wang et al. [4] report a steady-state peak/trough of 1.8× (789.8/435.6 ng/dL at week 12). This gap is *arithmetic, not structural*: over a 70-day interval, a terminal half-life of 20 d is 3.5 half-lives of decay, so peak/trough cannot fall below ∼11× on decay alone. Both targets *are* jointly reachable in the two-rate Bateman—a 73.8 d terminal half-life paired with a 27.5 h release rate hits $t_{\max} = 7$ d *and* peak/trough 1.80× simultaneously—but 73.8 d is 3.7× what Zhang measured. The blocker is that Zhang's measured terminal half-life and Wang's observed flatness cannot both be true; resolving it requires Wang's primary sampling schedule, not a model rewrite. This finding is consistent with Pastuszak et al. [5], who fit injectable TU with a one-compartment first-order-absorption model—the same two-rate-constant Bateman used here.

2.9.3 Testosterone propionate (TP)

TP remains uncalibrated. It retains a placeholder 2 h k_e (physiological clearance of free testosterone), which pins its $t_{\max}$ to 0.33 d against a published window of 1–2 d. Its *magnitude* was corrected by the clearance identity (Section 2.8), so steady-state C_{avg} is correct, but the first ∼24 h of the curve—where $t_{\max}$ lives—is wrong. The best available source is a textbook chapter [9] showing a simulated twice-weekly curve rather than fitted rate constants or raw single-dose data. A fabricated fit would be worse than the honest flag.

2.10 Estradiol ester derivations

2.10.1 Estradiol valerate (EV) and cypionate (EC)

EV and EC are calibrated against transfemscience.org's compiled tables [8], which aggregate published single-dose and steady-state estradiol PK. The calibration procedure:

1. k_e was fit so Equation 4 reproduces the published time-to-peak (2.1 d for EV, 3.1–4.3 d for EC).
2. k_a was fit so the modeled steady-state trough reproduces Table 10's 5 mg-weekly figures.
3. S was anchored to the same table's steady-state peak.

The resulting constants reproduce the published steady-state $C_{\max}$ and $C_{\min}$ to within 0.1% (Table 7). A key decision: single-dose and steady-state targets were *not simultaneously satisfiable* in the two-parameter model, and steady state was chosen on the reasoning that it is what users actually live at and what clinicians titrate on. The consequence is that single-dose $C_{\max}$ is a derived value, not an enforced target (Section 5.2).

2.10.2 Oral estradiol

Oral estradiol uses the same Bateman biexponential with fast rate constants. The absorption rate $k_a = 8.762\,\text{day}^{-1}$ ($t_{1/2} = 0.0791$ d) was fit so that $t_{\max} = 6.0$ h, the midpoint of the published 4–8 h window. The elimination rate $k_e = 1.386\,\text{day}^{-1}$ ($t_{1/2} = 12.0$ h) and $S = 1250$ were anchored to reproduce the steady-state average for 2 mg daily (90 pg/mL).

A coupling issue arises: S and k_a are coupled through the peak but not through the average (integrating the Bateman term gives AUC/dose $= D^{\text{eff}} S/k_e$, with no k_a in it). One (S, k_a) pair cannot hold both the published $t_{\max}$ and the published single-dose $C_{\max}$. Anchoring on C_{avg} keeps 2 mg daily at 90 pg/mL—unchanged by the k_a refit, and the quantity clinicians titrate on—at the cost of single-dose $C_{\max}$ reading 88 pg/mL rather than 60 pg/mL for 2 mg.

2.10.3 Transdermal estradiol

Transdermal estradiol uses the 24 hourly micro-dose construction (Equation 3) with $S = 155$, $F = 0.15$, and $k_a = \ln 2/1.5 = 0.462\,\text{day}^{-1}$. The stored k_e is unused because the micro-dose branch sets $k_e = 0$ (single-exponential decay).

2.11 Transdermal testosterone

Anchored to the AndroGel 1% prescribing information [7], Table 5 (Day 180 steady state, $n = 44$): $50\,\text{mg/day} \rightarrow C_{\text{avg}} = 555\,\text{ng/dL}$. The terminal half-life comes from the label's washout statement—levels return to baseline in 48–72 h, implying ∼5% residual at 60 h, giving $t_{1/2} = 0.5784\,\text{d}$. Transdermal $F = 0.10$ is stated outright in the label's §12.3: "Approximately 10% of the testosterone dose applied on the skin surface from AndroGel is absorbed into systemic circulation."

The scaling constant $S = 132.98$ is derived from the 50 mg anchor. Two limitations are documented rather than fitted around; both are in Section 5.1.

2.12 Oral testosterone undecanoate

Anchored to Swerdloff et al. [6] / the Jatenzo label: 237 mg twice daily $\rightarrow C_{\text{avg}} = 488.7\,\text{ng/dL}$, $t_{\max}$ 2–4 h. Both rate constants were solved jointly so that $t_{\max}$ lands at 3 h and the steady-state $C_{\max}/C_{\text{avg}}$ comes as close as the model permits to the reported 2.06. The resulting constants are $k_a = 7.702\,\text{day}^{-1}$ ($t_{1/2} = 0.09\,\text{d}$) and $k_e = 8.305\,\text{day}^{-1}$ ($t_{1/2} = 2.003\,\text{h}$), with $S = 13.57$.

2.13 Dynamic endogenous baseline

Endogenous hormone production is modeled as a time-varying baseline $C_{\text{base}}(t)$ that decays from an intact gonadal level to a suppressed level as HRT progresses. The model uses a saturating exponential taper:

$$C_{\text{base}}(t) = C_{\text{full}} + (C_{\text{supp}} - C_{\text{full}}) \cdot \left(1 - e^{-2.5\,p(t)}\right) \tag{8}$$

where $p(t) = \min(\text{weeks on HRT}/t_{\text{scale}}, 1)$ is the normalized progress, t_{scale} is user-configurable (4, 8, or 12 weeks), C_{full} is the intact baseline (100 pg/mL for estradiol, 600 ng/dL for testosterone), and C_{supp} is the suppressed baseline (5 pg/mL for estradiol, 15 ng/dL for testosterone). The factor 2.5 in the exponent ensures the suppression reaches ∼92% at $p = 1$. After 104 weeks (2 years), the baseline is treated as fully suppressed.

Anti-androgen override. When the user reports taking an anti-androgen (spironolactone, cyproterone, or a GnRH agonist), the testosterone baseline is set to the suppressed level immediately, reflecting the direct gonadal suppression these agents provide.

2.14 Blood test calibration

2.14.1 The calibration factor

Each blood test record stores a measured hormone level and the date the blood was drawn. The calibration factor is the ratio of measured to predicted concentration at the draw instant:

$$\alpha = \frac{C_{\text{measured}}}{C_{\text{predicted}}} \tag{9}$$

where $C_{\text{predicted}}$ is computed by running the PK model at the exact draw date *without* calibration applied (to avoid circularity: the calibration engine must never feed on its own output). The factor is clamped to $[0.2, 5.0]$ to exclude clinically implausible values. When multiple calibrated blood tests exist, the most recent one for the currently selected hormone type is used.

2.14.2 Application

α is applied as a multiplicative scalar to the entire model output (Equation 2). This is a pragmatic per-user analog of the inter-individual variability (IIV) terms in a formal mixed-effects population-PK model. The published literature reports substantial between-subject variability: Ichihara et al. [2] report a CV of 0.58 on V_d/F for TE, corresponding to an ∼8-fold individual range. The

clamp range $[0.2, 5.0]$ spans most of this distribution without admitting values that would indicate a data entry error.

2.14.3 Hormone-type gating

The calibration factor is only applied when the blood test's hormone type matches the currently selected medication. This prevents a testosterone draw from silently rescaling the estradiol curve, which would occur if the filter were absent.

2.15 Drug interactions: spironolactone

Spironolactone is a known CYP3A4 inducer that can modestly increase estradiol clearance. When the user has opted in to the spironolactone clearance adjustment, the elimination rate k_e for estrogen doses is scaled by a factor of 1.15 (15% faster clearance):

$$k_e' = k_e \cdot 1.15 \tag{10}$$

This adjustment is applied at the PK parameter resolution level, before the Bateman function is evaluated. It is not applied to testosterone doses, as the interaction is specific to estradiol metabolism.

Note that because $t_{\max}$ depends on k_e (Equation 4), this adjustment also shifts the peak time slightly earlier. The ester-level $t_{\max}$ values reported in Table 9 are computed without the spironolactone adjustment.

2.16 Aromatization estimation

For trans men on testosterone therapy, a small fraction of circulating testosterone is converted to estradiol via aromatase. The model estimates this conversion as:

$$E_2^{\text{arom}} = 0.002 \times T_{\text{ng/dL}} \quad \text{(pg/mL)} \tag{11}$$

i.e., a 0.2% conversion rate. This helps trans men monitor whether their estradiol is within the male reference range (10–40 pg/mL). The conversion factor is a population average; individual aromatase activity varies with body composition, age, and genetic factors.

2.17 Chart window and sampling resolution

The chart's visible time range and sampling step are derived from the ester's pharmacokinetic properties, not hardcoded. The rule is:

$$W = \text{clamp}(1.5 \cdot t_{\max},\ \text{lower: } \min(2\,\text{d},\ 2\tau),\ \text{upper: } 1.5\tau) \tag{12}$$

$$\text{step} = \max\big(\min(W/40,\ t_{\max}/8),\ 15\,\text{min}\big) \tag{13}$$

where W is the total window width, τ is the dosing interval, and the window is split 1:2 past:future. The $t_{\max}/8$ cap is an anti-aliasing guarantee: it ensures the sampling step never exceeds one-eighth of the time-to-peak, preventing the drawn curve from collapsing into a flat line when the step would otherwise land on the same phase of every dose cycle. The $W/40$ term keeps every ester at a comparable point count ($\sim$41) so curves read with the same density regardless of timescale.

When $t_{\max}$ is uncalibrated or non-finite (the transdermal routes, which have no meaningful $t_{\max}$ because $k_e = 0$), a fallback window of 2 d is used.

	slow exponential		fast				
Form	$t_{1/2}$ (d)	k_a (d^{-1})	k_e (d^{-1})	EWF	F	S	t_{max}
EV	2.6000	0.2667	0.7737	0.764	1.00	396.40	2.10 d
EC	6.4700	0.1071	0.4310	0.687	1.00	273.00	4.30 d
E2 oral	0.0791	8.7629	1.3859	1.000	0.05	1250.00	6.0 h
E2 cream	1.5000	0.4621	—	1.000	0.15	155.00	—
TC	3.8390	0.1806	1.2196	0.699	1.00	67.13	1.84 d
TE	6.6000	0.1050	1.9625	0.720	1.00	107.98	1.58 d
TP	1.0000	0.6931	8.3178	0.837	1.00	457.67	0.33 d
TU	20.0000	0.0347	0.3748	0.631	1.00	20.62	7.00 d
T oral (TU)	0.0900	7.7016	8.3053	0.631	1.00	13.57	3.0 h
T gel	0.5784	1.1984	—	1.000	0.10	132.98	—

Table 6: Constants as shipped, and the t_{max} each implies via Equation 4. The cream/gel routes have no t_{max} because they are not modeled with Equation 1; their k_e is stored but unused.

3 Encoded constants

Calibration status. Of the eight forms with a t_{max}, seven are calibrated against a primary or label source: EV and EC against compiled steady-state tables [8], oral E2 against the published 4–8 h window, TC against Bi et al. [1], TE against Ichihara et al. [2], TU against Aveed's FDA clinical pharmacology, and oral T against the Jatenzo label [6]. TP alone remains uncalibrated: its t_{max} of 0.33 d is known to be wrong against a published window of 1–2 d, and is not presented to users as a fact about the drug.

4 Results

4.1 Accuracy against anchors

Regimen	**Quantity**	**Model**	**Published**	**Error**	**Source**
EV 5 mg q7d	C_{max}	384	384	+0.1%	TFS Table 10
EV 5 mg q7d	C_{min}	142	142	+0.1%	TFS Table 10
EC 5 mg q7d	C_{max}	339	339	−0.1%	TFS Table 10
EC 5 mg q7d	C_{min}	262	262	−0.1%	TFS Table 10
TC 100 mg q7d	C_{avg}	550	549	+0.0%	Bi Table 2
T gel 50 mg/d	C_{avg}	555	555	+0.1%	AndroGel Table 5
T gel 75 mg/d	C_{avg}	833	601	+38.6%	AndroGel Table 5
T gel 100 mg/d	C_{avg}	1111	713	+55.8%	AndroGel Table 5
Oral TU 237 mg BID	C_{avg}	489	489	−0.0%	Jatenzo
Oral TU 237 mg BID	C_{max}	789	1008	−21.7%	Jatenzo

Table 7: Predicted versus published. Rows with near-zero error are the anchors themselves and demonstrate that the fit converged. The three coloured rows are the informative ones; both are explained in Section 5.1. Estradiol in pg/mL, testosterone in ng/dL.

Rows in Table 7 with 0.0% error demonstrate only that the fit converged—a fit agrees with its own target by construction. The three coloured rows are the informative ones.

4.2 Out-of-sample validation

The following comparisons were *not* used in any fit.

Check	Model	Published	Error	Source
TC single 200 mg C_{max} (exogenous)	997	872	+14.3%	200 mg IM lit. 1112 less 240 basal
TC 100 mg q7d C_{max} + basal	976	1150	−15.1%	Bi Figure 2, 100 mg panel
EV single 2 mg C_{max}	119	118	+1.0%	TFS Table 9
EC single 2 mg C_{max}	59	62	−5.1%	TFS Table 9

Table 8: Independent validation. The estradiol rows are the strongest evidence: constants fit purely to steady-state data reproduce single-dose peaks within 1% and 5%.

The two testosterone rows agree to about 15% in opposite directions, which is roughly the precision the source itself supports given Bi's reported RSEs. They are consistency checks, not tight validation.

The estradiol rows are the strongest evidence in the model: constants fit purely to steady-state data reproduce single-dose peaks within 1% and 5%. This is a non-trivial result—it demonstrates that the two-parameter Bateman, with steady-state-anchored constants, generalizes to the single-dose regime without refitting.

4.3 Time to peak

Form	Model t_{max}	Published	Source quality
EV	2.10 d	2.1 d	compiled tables
EC	4.30 d	3.1–4.3 d	compiled tables
E2 oral	6.0 h	4–8 h	secondary
TC	1.84 d	1.84 d	**primary** (Bi Table 2)
TE	1.58 d	1.7 ± 0.9 d	**primary** (Ichihara 2020)
TP	0.33 d	1–2 d	sparse (uncalibrated)
TU	7.00 d	median 7 d	**label** (Aveed FDA)
T oral	3.0 h	2–4 h	label (Jatenzo)

Table 9: Where the model puts the peak. Seven of eight forms with a t_{max} are calibrated against a primary or label source. TP alone remains uncalibrated: its t_{max} is known to be wrong and is not presented to users as fact.

4.4 Persona validation

Section-level agreement with published tables is necessary but not sufficient: those tables describe clean steady state on a fixed regimen, and real users miss doses, change dose, change route, and dose on intervals nobody studied. Eight synthetic users were run end to end through the same summation the app performs.

Persona	Regimen	What it tests, and the result
Maya, 24	4 mg EV IM q7d, 18 months, perfect adherence	Control. C_{max} 307, C_{min} 114 pg/mL — exactly 4/5 of the 5 mg source figures, as linearity requires. Pass.
Devon, 31	100 mg TC q7d → 200 mg q14d at month 6	Superposition across a regimen change. C_{avg} preserved at 790 ng/dL as it must be; trough falls 561 → 381 while peak rises 976 → 1339, and peak/trough widens 1.74× → 3.51×. The model reproduces the clinical trade-off and correctly refuses to reward the longer interval. Pass.

Persona	Regimen	What it tests, and the result
Priya, 47	2 mg oral E2 BID for 4 months, then 4 mg EV q7d	Route switch; historical doses must keep their own kinetics. They do — the oral segment retains $F = 0.05$ and the fast rates. Modeled BID swing is 1.37× at an unchanged C_{avg} of 180 pg/mL. Pass.
Sam, 19	50 mg/day T gel, misses ∼2 days/week	Skipped-dose handling. C_{avg} 637 ng/dL; missed days cost 20% of average and 63% of trough. Pass.
Rowan, 38	4 mg/day compounded E2 cream, onboarded in-app	Route pseudo-ester resolution. Cream dose correctly resolves to the transdermal pseudo-ester constants. Pass.
Kai, 52	237 mg oral TU BID, 10-day interruption	Washout and re-accumulation on the fastest route. Falls to the endogenous floor within a day; no phantom drug survives the gap; recovers to 100% of pre-gap within 2 days. Pass.
Jules, 28	100 mg TE IM q7d	TE steady-state. C_{max} 703, C_{min} 405, C_{avg} 566 ng/dL (exogenous). Consistent with TC via the clearance identity. t_{max} 1.58 d, inside Ichihara's observed 1.7 ± 0.9 d. Pass.
Nadia, 35	5 mg EC q10d — an interval nobody studied	Extrapolation outside the calibration envelope. Trough holds in band but sits 40% below the same dose at q7d. Structurally sound but inherits every assumption of the fit. Pass, with caveat.

All eight land inside their clinical reference bands. The persona suite tests properties that section-level accuracy tables cannot: linear scaling (Maya), superposition across regimen changes (Devon), route switching with per-dose kinetics (Priya), skipped-dose handling (Sam), route pseudo-ester resolution (Rowan), washout and re-accumulation (Kai), inter-ester consistency (Jules), and extrapolation outside the calibration envelope (Nadia).

5 Discussion

5.1 Structural limitations

These are quantified rather than fixed. Each needs either a source the model does not have or a structurally different approach.

5.1.1 Testosterone undecanoate peak/trough ratio

Over a 70-day interval, the model produces a peak/trough ratio of 8.1× against Wang et al.'s [4] observed 1.8×. This is not a structural deficiency of the two-rate Bateman: as noted in Section 2.9, both targets are jointly reachable with a 73.8 d terminal half-life, but that value is 3.7× what Zhang et al. [3] measured. The blocker is a *source conflict* between Zhang's terminal half-life and Wang's observed flatness, not the number of compartments. Pastuszak et al. [5] fit injectable TU with the same one-compartment first-order-absorption model used here, confirming the model family is appropriate.

The magnitude correction via the clearance identity improves TU's C_{avg} (496 ng/dL exogenous, against a published ∼500–600 total) but its C_{max} overshoots (1045 vs. 790 ng/dL observed),

because correcting magnitude cannot correct shape. The trade was taken knowingly: C_{avg} is what clinicians titrate on and what a user lives at.

5.1.2 Transdermal intraday ripple

The 24-hourly-micro-dose construction has no absorption phase and tops out at $C_{\max}/C_{\min} \approx 1.05$ for testosterone gel, against an observed 2.24. Reproducing 2.24 would require a terminal half-life near 1 h, which contradicts the label's own 48–72 h washout by an order of magnitude. The two cannot be satisfied together in this model. C_{avg} is exact.

5.1.3 Transdermal dose-response sublinearity

The AndroGel 1% label's dose-response is markedly sublinear: 50 mg → 555, 75 mg → 601, 100 mg → 713 ng/dL. The 1.62% label inverts outright: 81 mg C_{avg} 537 is *below* 60.75 mg C_{avg} 643.

This is a titration artifact: higher-dose arms are enriched for poor absorbers, because subjects were escalated precisely when they responded badly. It is selection confounding, not saturable pharmacokinetics, and a linear model should not be bent to reproduce it. S is anchored to the 50 mg arm—the standard starting dose and the least confounded—so predictions at 75 and 100 mg run high by 39% and 56% (Table 7).

Gel absorption also varies enormously between individuals. The calibration factor α is the intended remedy and is more important on this route than on any other.

5.1.4 Oral testosterone peak fluctuation

With $t_{\max}$ pinned to the reported 2–4 h window, the model's achievable $C_{\max}/C_{\text{avg}}$ tops out at 1.61 against a reported 2.06, so $C_{\max}$ reads 789 rather than 1008 ng/dL.

The reported ratio is itself inflated by mean-of-maxima bias: the published $C_{\max}$ is the mean of per-subject maxima, and $\mathbb{E}[\max] \geq \max[\mathbb{E}]$, with a reported SD of ±581 on a mean of 1008. Anchoring to C_{avg}—the label's own primary endpoint—and accepting a documented under-prediction of the peak was judged the better trade.

5.1.5 Testosterone propionate remains uncalibrated

TP retains a placeholder 2 h k_e, which pins its $t_{\max}$ to 0.33 d against a published window of 1–2 d. Its magnitude was corrected by the clearance identity, so steady-state C_{avg} is correct, but the peak arrives too early and the curvature between peak and trough is wrong. The best available source is a textbook chapter [9] showing a simulated curve, not fitted rate constants. A fabricated fit would be worse than the honest flag.

5.1.6 Model-wide simplifications

The model assumes:

- **Dose-linearity** with no saturable protein binding or SHBG dynamics.
- **A single well-mixed compartment** with constant clearance.
- **No covariates**: body composition, age, hepatic function, and concomitant medications (beyond the one spironolactone adjustment) are not represented.
- **No between-subject variability** in the population prediction. Predictions are population means, not individual forecasts. The per-user calibration factor α exists to close some of that gap once blood tests are available.

These are standard simplifications for a one-compartment model and are shared by the published population-PK models cited above [1, 2, 5], with the caveat that those models additionally include IIV terms and covariates (e.g., body weight and SHBG in Pastuszak's TU model).

5.2 The steady-state vs. single-dose anchor choice

For both estradiol esters and all four testosterone esters, the single-dose peak targets and the published steady-state data were *mutually unsatisfiable* in the two-parameter model. For estradiol the gap was large: holding the single-dose targets put steady state +110% (EV) and +206% (EC) above the published table.

Steady state was chosen, in every case, on the reasoning that it is what users actually live at and what clinicians titrate on. The consequence is that single-dose figures are derived values, not enforced targets. This is a modelling decision, not a derivation, and it has a cost: single-dose $C_{\max}$ for EV (119 pg/mL for 2 mg) and EC (59 pg/mL for 2 mg) are lower than some published single-dose figures, though they agree with the transfemscience Table 9 values to within 1% and 5% respectively (Table 8).

5.3 Source quality hierarchy

The model's constants rest on a hierarchy of source quality:

1. **Primary population-PK** (TC): Bi et al. [1] supply CL/F, V/F, and K_a with RSEs and bootstrap CIs. This is the strongest source in the model.
2. **Primary single-dose PK** (TE): Ichihara et al. [2] supply terminal half-life and per-subject absorption rates in the target population (transgender men, IM route).
3. **Regulatory labels** (T gel, oral T, TU $t_{\max}$): AndroGel [7] and Jatenzo [6] supply steady-state C_{avg}, $C_{\max}$, and $t_{\max}$ from FDA registration trials. Aveed's FDA clinical pharmacology supplies TU's median $t_{\max}$.
4. **High-quality secondary compilation** (EV, EC): transfemscience.org [8] aggregates published single-dose and steady-state estradiol data. Not a primary table, and flagged as such.
5. **Secondary literature** (TU terminal half-life): Zhang et al. [3] and Wang et al. [4] supply terminal half-life and steady-state peak/trough, respectively. These two sources are in conflict (Section 5.1.1).
6. **Nothing** (TP): beyond the clearance identity that ties TP's magnitude to TC, there is no primary source for TP's absorption rate. The placeholder is flagged.

5.4 The per-user calibration factor as a pragmatic IIV analog

The published population-PK models for testosterone esters all report substantial inter-individual variability. Ichihara et al. [2] report a CV of 0.58 on V_d/F for TE, and Bi et al. [1] report IIV terms on clearance and volume. A formal mixed-effects model would represent this variability with η terms on each parameter, drawn from a population distribution.

xT's calibration factor α is a simpler approach: a single scalar multiplier on the entire model output, derived from the user's own blood test. It cannot represent parameter-specific variability (e.g., a user with fast clearance but normal volume), and it cannot provide population-level confidence intervals. But it is practical for a consumer application: it requires no raw data access, no fitting infrastructure, and no statistical expertise from the user, and it absorbs the dominant source of individual-level error once a blood test is available.

The clamp range $[0.2, 5.0]$ spans most of the reported IIV distribution without admitting values that would indicate a data entry error. For routes with especially high between-subject variability—transdermal in particular—α is more important than on any other route.

6 Conclusion

We have presented the pharmacokinetic model underlying xT, a mobile application for hormone replacement therapy monitoring. The model employs a one-compartment Bateman biexponential with linear superposition of dose contributions, a dynamic endogenous baseline, and a per-user

calibration factor. Ten medication/route/form combinations are supported across estradiol and testosterone.

The model's key structural features are:

- A **conservation-of-mass identity** that constrains the scaling constant S for all injectable testosterone esters to depend only on k_e and clearance, eliminating S as a free parameter and enforcing mutual consistency across esters.
- An **analytic closed-form derivation** of S for testosterone cypionate from primary population-PK parameters, which anchors the identity.
- A **per-user calibration factor** α that serves as a pragmatic analog of inter-individual variability terms in formal mixed-effects models.
- A **dynamic endogenous baseline** that models gonadal suppression over time, separating exogenous from endogenous contributions.

The model achieves 0.0–0.1% accuracy against its anchors and 1–15% on out-of-sample checks. Seven of eight forms with a $t_{\max}$ are calibrated against primary or label sources; one (TP) is honestly flagged as uncalibrated. Structural limitations are quantified rather than hidden: TU's peak/trough ratio is a source conflict, not a structural deficiency; transdermal intraday ripple is under-modeled by construction; and oral peak fluctuation is under-predicted by an anchoring choice that prioritizes C_{avg}.

The model is honest about where its constants are primary-sourced, where they are secondary, and where they remain uncalibrated. This honesty— the explicit marking of what is known, what is estimated, and what is placeholder—is, in our view, the most important property of a pharmacokinetic model used in a consumer health context, where users may make medical decisions based on its output.

This paper describes a consumer tracking application's internal model. It is not clinical guidance, and its predictions are population estimates that should not be used to make dosing decisions without laboratory confirmation.

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